

Skill 10:

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/wait.h>
#include <ctype.h>
#include <limits.h>

/* =====
   Global state
   ===== */

char saved_directory[PATH_MAX];

/* =====
   Save current state
   ===== */

void save_state()
{
    if (getcwd(saved_directory, sizeof(saved_directory)) != NULL)
    {
        printf("State saved: %s\n", saved_directory);
    }
    else
    {
        perror("getcwd");
    }
}

/* =====
   Display current directory
   ===== */

void builtin_pwd()
```

```

void builtin_pwd()
{
    char current_directory[PATH_MAX];

    if (getcwd(current_directory, sizeof(current_directory)) != NULL)
    {
        printf("Current directory: %s\n", current_directory);
    }
    else
    {
        perror("pwd");
    }
}

/* =====
Cleanup resources
===== */

void cleanup()
{
    printf("Cleaning up resources...\n");

    /* No dynamically allocated resources
       are required in this example. */

    printf("Cleanup completed.\n");
}

/* =====
Exit request
===== */

void builtin_exit()
{

```

```
int name_length = equals - input;

if (name_length >= sizeof(name))
{
    printf("export: variable name too long\n");
    return;
}

strncpy(name, input, name_length);
name[name_length] = '\0';

/* Validate variable name */

if (!valid_name(name))
{
    printf("export: invalid variable name: %s\n", name);
    return;
}

/* Extract value */

strcpy(value, equals + 1);

/* Check existing variable */

char *old_value = getenv(name);

if (old_value != NULL)
{
    printf("Existing variable detected.\n");
    printf("Old value: %s\n", old_value);
}

/* Update environment variable */
```

```
int main()
{
    printf("=====\n");
    printf("PWD, EXIT AND EXPORT PRACTICAL\n");
    printf("=====\n");

    /* Retrieve current directory */

    printf("\n1. Current Directory:\n");
    builtin_pwd();

    /* Save state */

    printf("\n2. Saving State:\n");
    save_state();

    /* Export variable */

    printf("\n3. Exporting Environment Variable:\n");
    builtin_export("COURSE=CSE");

    /* Check variable */

    printf("\n4. Checking Environment Variable:\n");

    char *course = getenv("COURSE");

    if (course != NULL)
        printf("COURSE=%s\n", course);

    /* Existing variable test */
}
```

```
printf("\n5. Updating Existing Variable:\n");
builtin_export("COURSE=ComputerScience");

/* Invalid variable name */

printf("\n6. Testing Invalid Variable Name:\n");
builtin_export("123COURSE=Test");

/* Invalid syntax */

printf("\n7. Testing Invalid Export Syntax:\n");
builtin_export("INVALID");

/* Child process test */

printf("\n8. Testing Export in Child Process:\n");
test_child_environment();

/* Verify current directory */

printf("\n9. Verifying Current Directory:\n");
builtin_pwd();

/* Exit */

printf("\n10. Testing Exit Request:\n");
builtin_exit();

return 0;
```

```
}
```

```

bhanu@DESKTOP-DE32HQ8:~$ cd ~
mkdir -p ~/pwd_export
cd ~/pwd_export
bhanu@DESKTOP-DE32HQ8:~/pwd_export$ pwd
/home/bhanu/pwd_export
bhanu@DESKTOP-DE32HQ8:~/pwd_export$ nano pwd_export.c
bhanu@DESKTOP-DE32HQ8:~/pwd_export$ gcc pwd_export.c -o pwd_export
bhanu@DESKTOP-DE32HQ8:~/pwd_export$ ls
pwd_export  pwd_export.c
bhanu@DESKTOP-DE32HQ8:~/pwd_export$ ls
pwd_export  pwd_export.c
bhanu@DESKTOP-DE32HQ8:~/pwd_export$ ./pwd_export
=====
PWD, EXIT AND EXPORT PRACTICAL
=====

1. Current Directory:
Current directory: /home/bhanu/pwd_export

2. Saving State:
State saved: /home/bhanu/pwd_export

3. Exporting Environment Variable:
Export successful.
COURSE=CSE

4. Checking Environment Variable:
COURSE=CSE

5. Updating Existing Variable:
Existing variable detected.
Old value: CSE
Export successful.
COURSE=ComputerScience

6. Testing Invalid Variable Name:
export: invalid variable name: 123COURSE

7. Testing Invalid Export Syntax:
export: invalid syntax
Use: export NAME=VALUE

```

6. Testing Invalid Variable Name:
export: invalid variable name: 123COURSE

7. Testing Invalid Export Syntax:
export: invalid syntax
Use: export NAME=VALUE

8. Testing Export in Child Process:

Child process:
COURSE=ComputerScience
Child process completed.

9. Verifying Current Directory:
Current directory: /home/bhanu/pwd_export

10. Testing Exit Request:
Exit request received.
Cleaning up resources...
Cleanup completed.
Saving final state...
State saved: /home/bhanu/pwd_export
Shell exiting successfully.
bhanu@DESKTOP-DE32HQ8:~/pwd_export\$ |